

# Julia V.E. Kim

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## Education

### Massachusetts Institute of Technology

PH.D. OPERATIONS RESEARCH

Cambridge, US  
Sept. 2025–Present

- **Cumulative GPA:** 5.0/5.0
- **Research Interests:** AI agents for automated research, ML and optimisation, data-driven decision-making

### University of Cambridge, Trinity College

GRADUATE STUDIES IN MATHEMATICS

Cambridge, UK  
Oct. 2024–Jun. 2025

- **Scholarship:** Trinity Studentship in Mathematics
- **Thesis:** Applying Contrastive Learning to Stellar Spectra

### University of Toronto

BSc (HONOURS), PHYSICS, MATHEMATICS & STATISTICS

Toronto, CA  
Sep. 2019–Jun. 2024

- **Cumulative GPA:** 4.00/4.00, **Average:** 97.6%
- **Scholarship:** Rose Sheinin Award for *top female student* in science at UofT

## Professional Experience

### Data Science Intern

THE TRADE DESK: AI LAB

London, UK  
Jun. 2025–Aug. 2025

- Built end-to-end GPID validation and scoring pipeline in **PySpark** on **AWS S3**, processing 5.3M+ bidstream records for ad targeting
- Designed signal validation and data quality workflows for ad targeting, supporting marketplace quality research

### Astrophysics Machine Learning Researcher

UNIVERSITY OF TORONTO

Toronto, CA  
May 2024–May 2025

- Designed and trained convolutional neural networks in **PyTorch** to estimate stellar parameters from spectra, achieving  $R^2 > 0.95$
- Developed self-supervised contrastive learning models to align data from different modalities in a shared latent space

### Biophysics Machine Learning Researcher

UNIVERSITY OF TORONTO

Toronto, CA  
May 2023–May 2024

- Built a stochastic model in **TensorFlow** to simulate receptor signalling pathways with cross-talk and pleiotropy
- Applied information theory and statistical inference to investigate the impact of feedback and molecular noise on signalling specificity

### Dark Matter Particle Physics Researcher

SUPERCDCMS SNOLAB & UNIVERSITY OF TORONTO

Toronto, CA  
Sep. 2021–Apr. 2022

- Developed likelihood-based statistical models in **C++**, **RooFit** and **RooStats** to estimate detector sensitivity to dark matter signals
- Ran 1,000+ Monte Carlo simulations to construct test-statistic distributions and estimate signal upper limits at 95% confidence level

## Projects

### Personalised Instruction with Generative AI

PHD RESEARCH

MIT  
[GitHub](#)

- Developed a **Generative AI** system for automated educational content creation and personalised instruction at MIT
- Designed a multi-stage **LLM workflow** for automated assessment authoring and personalised video generation
- Designed content reuse and precomputation, projecting **46% lower generation time** from measured runtimes

### EgoBlind-RA: Risk-Adaptive Visual Assistance

MACHINE LEARNING COURSE PROJECT

MIT  
[GitHub](#)

- Trained a **CLIP** urgency classifier for visual assistance to blind users, achieving **0.905 test ROC-AUC**
- Fine-tuned **Kimi-VL** with **LoRA**, reducing composite loss by **38%** versus baseline on 1,283 test examples
- Evaluated **SFT** and **DPO** training strategies, analysing preference-pair design and robustness to routing errors

## Skills

**Programming:** Python, SQL, R, MATLAB, Julia, Bash

**ML & Data:** PyTorch, TensorFlow, scikit-learn, Hugging Face, PySpark, pandas, NumPy

**Tools:** Git, AWS S3, SLURM

**Languages:** English and French (native), Turkish (intermediate; coursework at **Harvard**)

## Publications

1. Dimitris Bertsimas, **Julia Kim**, and Romain Puech. “An Offline Architecture for Assessment-Driven Remediation Videos.” *Manuscript in preparation*, 2026.
2. **Julia V.E. Kim** & John R. Percy, “Pulsating Red Giants in a Globular Cluster:  $\omega$  Centauri”, 2022, JAAVSO, 50, 2. [Link](#)
3. Natasha Leclerc, **Julia V.E. Kim** et al. ‘Ba Depletion in Coastal Waters Points to Planktonic Phytoplankton Intensification’. (in progress)
4. **Julia V.E. Kim**. ‘Patterns and Trends in Endangered Species Listing Heavily Biased Towards Imperiled Vertebrates with High Utility Value to Humans.’ Featured in Alexander, R. (2024) *Telling Stories with Data with Applications in R*. Milton: CRC Press LLC. [Link](#)
5. **Julia V.E. Kim**. ‘Significant Declines in Standardised Test Scores due to COVID-19 School Closures Disproportionately Affect Vulnerable Students.’ Featured in Alexander, R. (2024) *Telling Stories with Data with Applications in R*. Milton: CRC Press LLC. [Link](#)

## Leadership

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|-------------------------------------------------------------------------------|--------------------|
| <b>MIT OGE</b> , Graduate Community Fellow                                    | Aug. 2026–Present  |
| <b>MIT ORC</b> , Student Seminar Organiser                                    | Aug. 2026–Present  |
| <b>MIT Student INFORMS Chapter</b> , Vice President                           | Sep. 2025–Present  |
| <b>Trinity College Cambridge</b> , APPD Programme Assistant                   | Oct. 2024–May 2025 |
| <b>Girls in STEAM</b> , Vice President of Content                             | Jan. 2023–May 2024 |
| <b>University of Toronto Machine Intelligence Team</b> , Academic Coordinator | Jan. 2023–May 2024 |

## Teaching

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|-----------------------------------------------------------------------|---------------------|
| <b>MIT Universal AI</b> , Course Content Development                  | May 2026–Present    |
| <b>University of Toronto</b> , Teaching Assistant, MAT137Y1: Calculus | Sept. 2020–May 2021 |

## Selected Awards & Honours

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|------------------------------------------------------------------------|-----------|
| <b>Jonathan Whitney Fund</b> , MIT                                     | 2026      |
| <b>Rose Sheinin Award</b> , University of Toronto                      | Jun. 2024 |
| <b>Trinity Studentship in Mathematics</b> , University of Cambridge    | Apr. 2024 |
| <b>Stirling Prize Graduate Fellowship</b> , Yale University (declined) | Feb. 2024 |
| <b>NSERC USRA and University of Toronto Chair Scholar</b>              | May 2022  |